

Group and Representation Theory

Chapter 2. Lie Groups

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Lie Groups

Lie Group

A **Lie group** G is a differentiable manifold which is endowed with a group structure such that the group operations

① $\cdot : G \times G \rightarrow G,$

② $^{-1} : G \rightarrow G$

are differentiable.

Examples. Orthogonal group $O(n)$, special linear group $SO(n)$, special Unitary group $SU(n)$, Lorentz group $O(1,3) = \{M \in GL(4, \mathbb{R}) | M\eta M^t = 1\}$ are Lie groups.

Generators

Let G be a **Lie group**, where group elements $g \in G$ depend **smoothly** on a set of continuous parameters α

- It is natural to set

$$g(\alpha = 0) = e \quad (1)$$

- Rep of G , denoted D , will generally be a function of α , and

$$D(\alpha = 0) = \mathbf{1} \quad (2)$$

Let us expand a unitary rep $D(\alpha)$ around the identity

$$D(d\alpha) = 1 + id\alpha_a X_a + \cdots$$

Generators

Generators

$$X_a \equiv -i \frac{\partial}{\partial \alpha_a} D(\alpha)|_{\alpha=0} \quad (3)$$

The X_a ($a = 1, \dots, N$) are called the **generators of the group**.

Unitarity of D implies X_a is **Hermitian** :

$$\begin{aligned} D^\dagger(\alpha) D(\alpha) &= (1 - id\alpha_a X_a^\dagger + \dots)(1 + id\alpha_a X_a + \dots) \\ &= 1 + id\alpha_a (X_a - X_a^\dagger) + \mathcal{O}(d\alpha^2) \\ &= 1 \end{aligned}$$

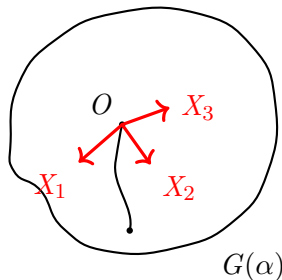
Actually, generators can be defined without mentioning representations (see [1])

Generators

Generators indicate specific directions in the group space : $X_a d\alpha_a$

→ By 'Stacking' infinitesimal elements, we can eventually reach $D(\alpha)$ even if α is **finite**!

$$D(\alpha) = \lim_{k \rightarrow \infty} \left(1 + i\alpha_a \frac{X_a}{k} \right)^k = e^{i\alpha_a X_a} \quad (4)$$



This rep for group elements, $D(\alpha)$, is known as the **exponential parameterization**

Lie Algebras

Suppose there are two group elements corresponding to different directions :

$$U_1 = \exp(i\alpha_a X_a), \quad U_2 = \exp(i\beta_b X_b) \quad (5)$$

Since $U_1 U_2$ is also a group element,

$$U_1 U_2 = \exp(i\delta_c X_c) \quad (6)$$

Let us expand both sides of the equality and compare the coefficients :

$$\begin{aligned} & \left(1 + i\alpha_a X_a + \frac{1}{2} (i\alpha_a X_a)^2 + \cdots \right) \left(1 + i\beta_b X_b + \frac{1}{2} (i\beta_b X_b)^2 + \cdots \right) \\ &= \left(1 + i\delta_c X_c + \frac{1}{2} (i\delta_c X_c)^2 + \cdots \right) \end{aligned}$$

Lie Algebras

$$\begin{aligned} & 1 + i\alpha_a X_a + i\beta_b X_b \\ & \quad - \alpha_a \beta_b X_a X_b - \frac{1}{2}(\alpha_a X_a)^2 - \frac{1}{2}(\beta_b X_b)^2 + \frac{1}{2}(\alpha_a X_a + \beta_b X_b)^2 + \cdots \\ & = 1 + i\alpha_a X_a + i\beta_b X_b - \frac{1}{2}[\alpha_a X_a, \beta_b X_b] + \cdots \\ & = 1 + i\delta_c X_c + \cdots \end{aligned}$$

Comparing terms, we conclude that

$$[\alpha_a X_a, \beta_b X_b] = -2i(\delta_c - \alpha_c - \beta_c)X_c \equiv i\gamma_c X_c \quad (7)$$

So if we require closure of G , generators must satisfy special commutation relations, known as the **Lie algebra**

Lie Algebras

Suppose that $[X_a, X_b] = if_{abc}X_c$, then we identify f_{abc} as

$$\alpha_a\beta_b[X_a, X_b] = i\alpha_a\beta_b f_{abc} = i\gamma_c$$

Also we have $f_{abc} = -f_{bac}$, since the commutator is anti-symmetric

Lie Algebra

Let X_a be generators of the Lie group G . The **Lie algebra** of the group is the commutation relation of generators such that

$$[X_a, X_b] = if_{abc}X_c \tag{8}$$

where f_{abc} are the **structure constants**.

Surprisingly, if we specify Lie algebra, group multiplication laws are fixed!

Structure Constants

- 1 For a fixed basis, the structure constants are independent of the reps
- 2 If X_a hermitian,

$$\begin{aligned}[X_a, X_b]^\dagger &= -if_{abc}^* X_c^\dagger \\ &= [X_b, X_a] = -if_{abc} X_c\end{aligned}$$

Therefore for unitary reps, structure constants are **real**

- 3 **The Jacobi identity**

$$[X_a, [X_b, X_c]] + (\text{Cyclic permutations}) = 0 \quad (9)$$

holds, and in terms of the structure constants

$$f_{bcd}f_{ade} + f_{abd}f_{cde} + f_{cad}f_{bde} = 0 \quad (10)$$

Formal Definition of Lie Algebra

For mathematical purposes, we present the formal definition of Lie algebra

Lie Algebra

A **finite-dimensional real^a/complex Lie algebra** is a finite-dimensional real/complex vector space $\mathfrak{g}$, together with a map (commutator) $[\cdot, \cdot] : \mathfrak{g} \times \mathfrak{g} \rightarrow \mathfrak{g}$ such that :

- 1 $[\cdot, \cdot]$ is bilinear.
- 2 $[\cdot, \cdot]$ is **skew-symmetric** :

$$[X, Y] = -[Y, X] \quad \forall X, Y \in \mathfrak{g}$$

- 3 The **Jacobi identity** holds :

$$[X, [Y, Z]] + (\text{Cyclic permutations}) = 0.$$

^aMathematicians often use anti-Hermitian generators, $t_a = -iX_a$, $[t_a, t_b] = f_{abc}t_c$

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The Adjoint Representation

Define a set of matrices T_a as ($a = 1 \cdots N$)

$$[T_a]_{bc} \equiv -if_{abc} \quad (11)$$

Then equation [10] reads

$$[T_a, T_b] = if_{abc} T_c \quad (12)$$

Therefore **matrices** T_a ($a = 1 \cdots N$) satisfy the commutation relation of generators

The Adjoint Representation

Matrices $[T_a] \in M_{N \times N}$ form an N -dimensional representation of the Lie algebra, called the **adjoint representation**.

The Adjoint Representation

Can we define a suitable '**scalar product**' between generators T_a and T_b ?

① $\langle T_a, T_b \rangle = T_a^T T_b \dots?$

scalar product must satisfy

$$\langle T_a, T_b \rangle = \langle T_b, T_a \rangle$$

② **Traces** $\text{tr}(T_a T_b)$ are cyclic :

$$\text{tr}(T_a T_b) = \text{tr}(T_b T_a)$$

Also linear :

$$\text{tr}(T_a(xT_b)) = x \text{tr}(T_a T_b), \quad \text{tr}(T_a(T_b + T_c)) = \text{tr}(T_a T_b) + \text{tr}(T_a T_c)$$

Therefore $\text{tr}(T_a T_b)$ is a suitable scalar product for our purposes

The Adjoint Representation

Note that generally T_a, T_b are not orthogonal

→ Can we transform the generators X_a to obtain an orthogonal basis?

Regard $\text{tr}(T_a T_b)$ as a real symmetric matrix (indices a, b)

$$\begin{pmatrix} \text{tr}(T_1 T_1) & \text{tr}(T_1 T_2) & \cdots & \text{tr}(T_1 T_N) \\ \text{tr}(T_2 T_1) & \ddots & & \\ \vdots & & \ddots & \\ \text{tr}(T_n T_1) & & & \text{tr}(T_N T_N) \end{pmatrix} \quad (13)$$

Restatement : Can we put this matrix into canonical form, by suitable linear transformation on generators X_a ?

The Adjoint Representation

Let's make a linear transformation on generators X_a

$$X_a \rightarrow X'_a = L_{ab} X_b$$

We want our commutation relation to be unchanged, so

$$\begin{aligned} [X'_a, X'_b] &= [L_{ad} X_d, L_{be} X_e] = i L_{ad} L_{be} f_{dec} X_c \\ &= i L_{ad} L_{be} f_{deg} L_{gc}^{-1} X'_c \\ &= i f'_{abc} X'_c \end{aligned}$$

Therefore structure constants transform as

$$f_{abc} \rightarrow f'_{abc} = L_{ad} L_{be} f_{deg} L_{gc}^{-1} \quad (14)$$

The Adjoint Representation

Note its resemblance to 'index' raising and lowering rules, so we could have just written

$$f_{ab}^c = L^{ad} L^{be} f_{de}^g L_{gc}, \quad L_{ab} \equiv (L^{ab})^{-1}$$

Therefore, matrices $[T_a]_{bc}$ transform as

$$[T_a] \rightarrow [T_a]' = L_{ad} L_{be} [T_d]_{eg} L_{gc}^{-1} \quad (15)$$

$$= L_{ad} L [T_d] L^{-1} \quad (16)$$

Therefore linear transformation on X_a induces

Linear transformation on generator indices + Similarity transformation

The Adjoint Representation

Since traces are invariant under similarity transformations,

$$\mathrm{tr}(T_a T_b) \rightarrow \mathrm{tr}(T'_a T'_b) = L_{ac} L_{bd} \mathrm{tr}(T_c T_d) \quad (17)$$

Thus we can **diagonalize** our 'trace matrix' $\mathrm{tr}(T_a T_b)$!

$$\mathrm{tr}(T_a T_b) = k^a \delta_{ab} \quad (\text{No sum}) \quad (18)$$

We still have the freedom to **rescale** the generators, so we can let k^a have *any* value (without changing its sign)!

Important Assumption

For convenience, we will assume that all k_a s are **positive**. This class of algebras is called the **compact Lie algebras**.^a

^aFor $k^a < 0$, see appendix [30]

The Adjoint Representation

Rescale X_a so that

$$\mathrm{tr}(T_a T_b) = \lambda \delta_{ab}, \quad \lambda > 0 \quad (19)$$

$$\begin{aligned} \mathrm{tr}([T_a, T_b] T_c) &= i f_{abd} \mathrm{tr}(T_d T_c) \\ &= i \lambda f_{abc} \end{aligned}$$

Since the trace is cyclic, we can deduce

$$\begin{aligned} \mathrm{tr}([T_a, T_b] T_c) &= \mathrm{tr}(T_a T_b T_c - T_b T_a T_c) \\ &= \mathrm{tr}(T_b T_c T_a - T_c T_b T_a) = \mathrm{tr}([T_b, T_c] T_a) \\ \therefore f_{abc} &= f_{bca} = f_{cab} \end{aligned} \quad (20)$$

So the structure constants f_{abc} are **completely antisymmetric** in this basis!

The Adjoint Representation

In this basis, the adjoint representation is **unitary**, since

$$[T_a]_{bc} = -if_{abc} = -\lambda^{-1} \text{tr}([T_a, T_b] T_c) \quad (21)$$

implies that $[T_a]$ is purely imaginary, and antisymmetric, i.e. **hermitian**

Mathematical description

The adjoint representation acts on $\mathfrak{g}$ as

$$X_a : X_b \mapsto [X_a, X_b] = if_{abc} X_c \quad (22)$$

So the adjoint rep is essentially $X \mapsto \text{ad}_X : \mathfrak{g} \rightarrow \mathfrak{g}$. Of course, the matrix elements are realized as $(\text{ad } X_a)_{bc} = [T_a]_{bc} = -if_{abc}$.

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Simple Algebras

Ideal

An **ideal**^a of a Lie algebra $\mathfrak{g}$ is a subspace $\mathfrak{h}$ spanned by generators X such that

$$[Y, X] \in \mathfrak{h} \quad \forall Y \in \mathfrak{g}.$$

^aGeorgi uses 'invariant subalgebra'. In this talk, we'll use standard mathematical terminology.

Let $h = e^{iX}$, $g = e^{iY}$ ($X \in \mathfrak{h}$, $Y \in \mathfrak{g}$)

$$g^{-1}hg = e^{iX'}$$

$$X' = e^{-iY} X e^{iY} = X - i \underbrace{[Y, X]}_{\in \mathfrak{h}} - \frac{1}{2} \underbrace{[Y, [Y, X]]}_{\in \mathfrak{h}} + \dots$$

Therefore the subgroup generated by the ideal $\mathfrak{h}$ is **normal**

Simple Algebras

Simple Algebra

An algebra which has no nontrivial proper ideal is called **simple**.

Theorem

The adjoint representation of a compact simple Lie algebra is irreducible.

proof. Assume that the theorem is false. Let $W \in \mathfrak{g}$ be a proper subspace, invariant under the adjoint action. But this implies

$$[X, W] \subset W, \quad \forall X \in \mathfrak{g},$$

which implies W is a nonzero proper ideal. Therefore $\mathfrak{g}$ is not simple. A contradiction. $\square$

Simple Algebras

We're interested in a special kind of subalgebra :

$U(1)$ factor of the group

Let X be a generator of a Lie algebra such that

$$[X, Y] = 0 \quad \forall Y \in \mathfrak{g}.$$

Then this generator is called a $U(1)$ **factor** of the group, or **abelian invariant subalgebra**.

$U(1)$ factors do not appear in the structure constants, and actually correspond to $k^a = 0$ direction in [18]

Lie algebras without any $U(1)$ factor are called **semisimple**

Semisimple Algebras

A **semisimple algebra** is useful, since

- ① It can be built by putting simple algebras together
- ② Every generator has a non-zero commutator with other generators
- ③ Every generator is a linear combination of commutators of generators

Essentially, structure constants contain all information about the algebra if it is semisimple

Assumption 2

Unless otherwise stated, all Lie algebras are assumed to be semisimple.

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States and Operators

Let $|i\rangle$ be vectors in a Hilbert space $\mathcal{H}$. Then the group elements $e^{i\alpha_a X_a}$ transform the vectors as

$$|i\rangle \rightarrow |i'\rangle = e^{i\alpha_a X_a} |i\rangle, \quad \langle i| \rightarrow \langle i'| = \langle i| e^{-i\alpha_a X_a} \quad (23)$$

Where the operators transform as

$$O \rightarrow O' = e^{-i\alpha_a X_a} O e^{i\alpha_a X_a} \quad (24)$$

Under infinitesimal transformation,

$$\begin{aligned} \delta |i\rangle &= (1 + i\alpha_a X_a - 1) |i\rangle = i\alpha_a X_a |i\rangle \\ \delta O &= (1 - i\alpha_a X_a) O (1 + i\alpha_a X_a) - O \\ &= -i[\alpha_a X_a, O] \end{aligned}$$

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Noncompact Lie Algebras

Let's consider one Lorentz algebra generator K_1 (see ref [2]), defined as

$$K_1 = i \begin{pmatrix} 0 & -1 & & \\ -1 & 0 & & \\ & & 0 & \\ & & & 0 \end{pmatrix}$$

Then we see $\text{tr}(K_1 K_1) = -2$, so Lorentz algebra itself is **noncompact**

Theorem

There is no nontrivial finite-dimensional unitary rep of a noncompact simple real Lie algebra.

For definitions and proof, see ref [3].

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References

- ① M. Nakahara, *Geometry, Topology and Physics*
- ② M. Schwartz, *Quantum Field Theory and the Standard Model*
- ③ Hall, *Lie Groups, Lie Algebras and Representations*